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RESEARCH ARTICLE



Integrated continuous biomanufacturing platform with ATF perfusion and one column chromatography operation for optimum resin utilization and productivity

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ABSTRACT

A new integrated continuous biomanufacturing platform for continuous production of antibodies at fixed cell volumes and cell concentrations for extended periods with immediate capture is presented. Upstream antibody production has reached technological maturity, however, the bottleneck for continuous biomanufacturing remains the efficient and cost-effective capture of therapeutic antibodies in an initial chromatography step. In this study, the first successful attempt at using one-column continuous chromatography (OCC) for the continuous capture of therapeutic antibodies produced through alternating tangential flow perfusion is presented. By performing upstream media optimizations, the upstream perfusion rate was reduced to one vessel volume per day (vv/d), increasing antibody titer and reducing the volume of perfusate. In addition, process improvements were performed to increase productivity by 80% over previously reported values. In addition, a real-time method for evaluating column performance to make column switching decisions was developed. This improved productivity coupled with the use of a single-column improved process monitoring and control in OCC compared to multi-column systems. This approach is the first report on using a single column for the implementation of an integrated continuous biomanufacturing platform and offers a cost-effective and flexible platform process for the manufacture of therapeutic proteins.

KEYWORDS

Antibody production; continuous chromatography; dynamic binding capacity; integrated continuous biomanufacturing; protein A

Introduction

Continuous manufacturing offers significant advantages over batch processes in terms of equipment footprint, batch to batch consistency, high productivities, and reduced capital cost.^[1–5] Despite these advantages, the biotechnology industry has been slow to adopt changes to the traditional batch and fed-batch processes because of concerns over production interruption due to increased complexity and stringent regulation.^[3,6,7] This explains why only minor changes have been implemented over multiple decades even with considerable technological improvements.

In the near future, due to the constantly evolving business environment, the renewed interest in single-use systems and diverse regulatory environments, therapeutic antibody manufacturers would have to adopt significant changes to contain cost, offer flexibility, maintain consistent product quality, and easily adjust to the rapidly and frequently changing global supply and demand environments.^[8] One approach would be to develop a cost-effective platform process for the production and capture of a diverse range of therapeutic proteins.

Historically, continuous cell cultures have been used for the production of therapeutic proteins, especially for the production of labile products such as Cerezyme[®] (Genzyme) and Kogenate[®] (Bayer).^[3,7] The motivation for using continuous cell cultures was to deal with the highly unstable

nature of the therapeutic proteins. These labile products are harvested and removed from the bioreactor in continuous cultures to prevent degradation. The integration of these continuous upstream processes with a continuous downstream process to produce a fully integrated continuous biomanufacturing system has still not been realized commercially.

Continuous purification in the pharmaceutical industry was pioneered by the simulated moving bed (SMB) technology using six columns and has progressed to the twin column system by ChromaCon[®]. Figure 1 summarizes a few examples on the evolution of continuous purification.

Recent advances in upstream–downstream process integration have proposed systems consisting of a continuous upstream coupled with a multicolumn continuous downstream process. Examples of these include the SMB with four to six columns,^[9,10] sequential multicolumn chromatography (SMCC) with four columns,^[11,12] periodic counter-current chromatography (PCC) with three columns^[4,13], and the twin-column chromatography with two columns.^[14]

The multicolumn approach is linked to the higher perfusion rates of three to five reactor volumes per day.^[13] By performing innovative media formulations, the current approach has been optimized to one reactor volume/day while keeping nutrient and wastes under control. This alone represents a multifold reduction in media use (cell culture

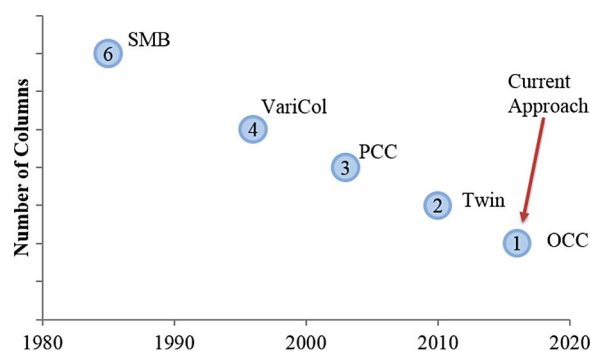


Figure 1. Evolution of number of columns in continuous purification. Current approach of single-column chromatography is 5 years ahead of industry prediction.

media ~\$70/L). The lower perfusion rates upstream result in smaller volumes of perfused material with higher concentration of antibodies. By implication, downstream operations require fewer purification cycles and smaller column sizes. Hence cost savings both upstream and downstream.

In this study, the first successful integrated continuous biomanufacturing platform comprised of a perfusion culture coupled with an OCC for the production of therapeutic antibodies with consistent product quality is presented. By performing process and media optimization strategies, the rate of perfusion was matched to the purification rate. The key improvements in our process over recent attempts are improved perfusion parameters to reduce media use (a significant cost driver in cell culture), production of higher antibody concentration perfusate because of lower perfusion rates, and optimization of resin dynamic binding capacity by low flow rates leading to a 20-fold reduction in protein A use. Moreover, the use of a single-column downstream capture step is technically very practical and can be easily implemented across companies with diverse plant and equipment features better than other technologies such as SMB, SMCC, and PCC.

It could be argued that the noncontinuous load proposed in the OCC approach is not continuous. However, as shown in Figure 2, even multicolumn systems operate as cyclic continuous processes at the level of individual columns. The justifications for multicolumn operation have been an increase in productivity and also as a method to process high flow effluent streams from continuous cell cultures. Moreover, even with multicolumn operations, a surge tank is still required to provide a buffer during process interruptions such as expired column replacements. Multiple downstream columns increase process complexity, equipment costs, and failure modes.

The OCC proposed in this work seeks to address these issues by eliminating the need for multiple columns. To achieve this, a combination of upstream process optimization, to reduce perfusion rates, and downstream process improvements, to improve productivity, will be combined to develop a simple, robust, and flexible integrated upstream/one-column downstream process for the manufacture of therapeutic antibodies.

Materials and methods

Cell culture

Perfusion mode cell culture was conducted in a 3 L glass bioreactor with 2 L working volume using an ATF2 system (Refine Technology, Pine Brook, NJ) equipped with a 1,000 cm² (3/4" radius, 65 cm length) 0.2-μm hollow fiber filter (Spectrum Laboratories, Rancho Dominguez, CA). Dissolved oxygen was controlled using O₂ gas through a sintered sparger. Cell density (Life Technologies, Grand Island, NY), nutrients, and waste were monitored by offline measurements using a NOVA Bioanalyzer 400 (Nova Biomedical, Waltham, MA). Chemically defined medium was modified through internal experiments to optimize bioreactor performance. Proprietary CHO cell lines producing biosimilars were used. The cells were seeded at 0.5×10^6 cells/mL. Cell growth,

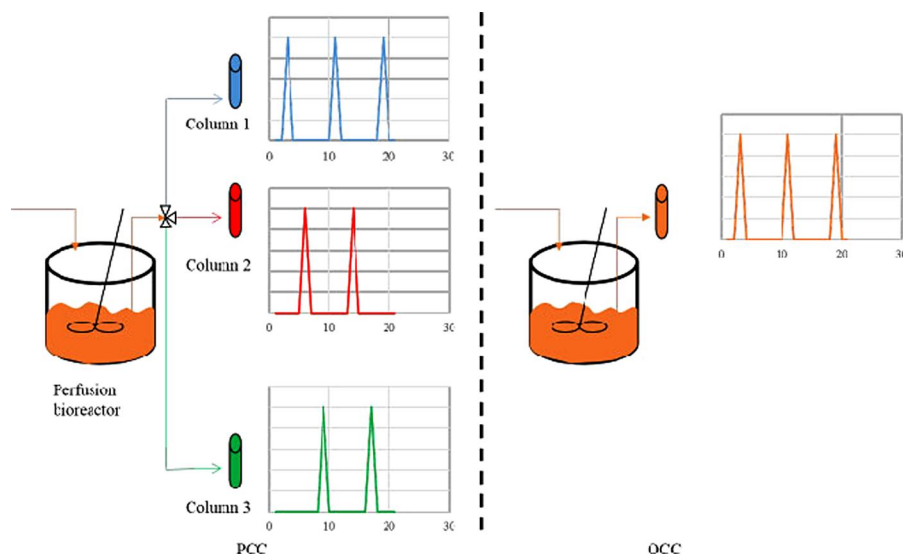


Figure 2. Comparison of the PCC and OCC process operations. The nature of capture–elute in affinity chromatography results in a cyclic continuous process in both scenarios. The number of columns are used to ensure timely processing of effluent bioreactor fluid. Note: OCC, one-column continuous chromatography; PCC, periodic counter-current chromatography.

lactate production, and ammonia production were controlled by perfusion rate, and nutrient levels were controlled by spiking media with additional nutrients from a concentrated broth. The process was optimized to an average perfusion rate of one reactor volume per day. pH was controlled around 7.2 with sodium bicarbonate and CO₂ gas. Dissolved oxygen (DO) was kept above 40%. The harvest was loaded directly onto a single MabSelect protein A column (GE Healthcare, Pittsburgh, PA) without additional clarification.

Downstream

Antibody capture was done with a 5 mL prepacked MabSelect column connected to a Quantasep100 unit (Sepragen, Hayward, CA). Each purification cycle consisted of an equilibration, a load, a wash, an elution, and a regeneration step. Although the unit did not provide a sterile operation, Sepragen now offers the same units with single-use flow lines which would offer sterile operation. This feature can be very useful during current good manufacturing practices (cGMP) manufacturing.

Method synchronization

Using the 2-L bioreactor, the perfusion rate was optimized during the antibody production phase to 1 vv/day corresponding to 2,000 mL of spent media every 24 hr. A single-column operation was optimized to 35 min with a 50 mL load of perfusate per cycle. A 20 min cleaning/sanitization cycle was conducted after every 20 cycles. This resulted in 40 complete cycles every 24 hr or 2,000 mL of perfusate every 24 hr, equivalent to the perfusion rate. The synchronization of perfusion rates and antibody capture rates while maintaining glucose and lactate levels at desired rates was thus achieved.

Analytical methods

Titer was quantified using ELISA. A 100 μ L sample of Anti-CHO-Alkaline Phosphatase was used to coat a flat-bottom 96-well plate for 1 hr at 37°C. The wells were washed thrice with 200 μ L of wash concentrate (Tris buffered saline with preservative). The plates were blocked with 200 μ L of blocking buffer at room temperature for 3 hr. To each well was added 100 μ L of standards, samples, and control after corresponding dilutions. The plate was incubated at 37°C for 1 hr. The samples were washed thrice with 200 μ L of wash concentrate, and 100 μ L of the secondary antibody was added and incubated at 37°C for an additional 1.5 hr. The plates were washed thrice with 200 μ L of wash concentrate. An amount of 100 μ L of p-nitrophenyl phosphate in a diethanolamine buffer with preservative (PNPP substrate) was added and incubated in the dark for 1.5 hr. The absorbance at 405/492 nm was read.

Host cell protein (HCP) in the bioreactor sample and purified sample was quantified by ELISA using commercial kits (Cygnus Technologies, Plainville, MA). The standard provided by the manufacturer was used to build the calibration curve between 2 and 100 ng/mL.

OCC approach

A single-protein A cycle involves an equilibration, a load, a wash and an elution step. The OCC approach is summarized in Figure 3. While the wash, elute and equilibration steps are progressing, enough perfusate for a single chromatographic cycle load step accumulates in the holding surge container. In fact, a slightly larger amount than the perfused harvest is loaded: In each cycle lasting 35 min, 48.6 mL of harvest is perfused from the reactor but 50 mL of perfusate is loaded into the column. However, once every 20 cycles (700 min), there is a sanitization/regeneration step that lasts 20 min. The extra perfusate build up during this sanitization/regeneration step compensates for the extra material loaded during each purification cycle.

The amount of material in the surge container can be minimized to near zero between each cycle during future process optimization approaches. In practice, however, a 1 L surge bottle holding about 200 mL of material was used since the process could not be supervised overnight. The 200 mL volume was chosen to ensure appropriate mixing by incoming material, so as to avoid stale perfusate over time which might result in antibody degradation and/or aggregation, contaminating the pooled sample and requiring additional purification downstream. This larger surge bottle was to provide a buffer in case the purification system paused overnight (a slight leak on the Quantasep units is detected by a leak detector causing a pause in the system).

The unique feature of this approach is the perfect synchronization with the perfusion rate upstream. A single chromatographic cycle was optimized to 35 min with the flow rate for loading only half of those for the other steps. The low loading flow rates increased dynamic binding capacities to levels very close to the static binding capacities, a feature that will be useful as future process optimizations of perfusion cell cultures will increase bioreactor titers. In addition to this, the OCC approach offers other very useful advantages as presented in the following section.

Lower protein A use

The key feature with the OCC approach is the use of a single-protein A column compared to SMB and PCC. For example, 1,000 L of culture produced by perfusion over 30 days would require only at most 1 L of protein A compared to at least 200 L in batch and fed-batch processes. This represents a 200-fold savings in protein A media alone (protein A media is ~\$75,000/L). It is possible to reduce the column size and run multiple cycles. However, to use the same 1 L of resin need to process 1,000 L in the OCC approach, at the same resin capacity, about 60–100 cycles will be required from the end of the batch or fed-batch process. This could add weeks to the overall process time and cause significant delays in product release.

Facility flexibility

The proposed OCC platform offers significant flexibility to organizations of varying sizes and equipment setup.

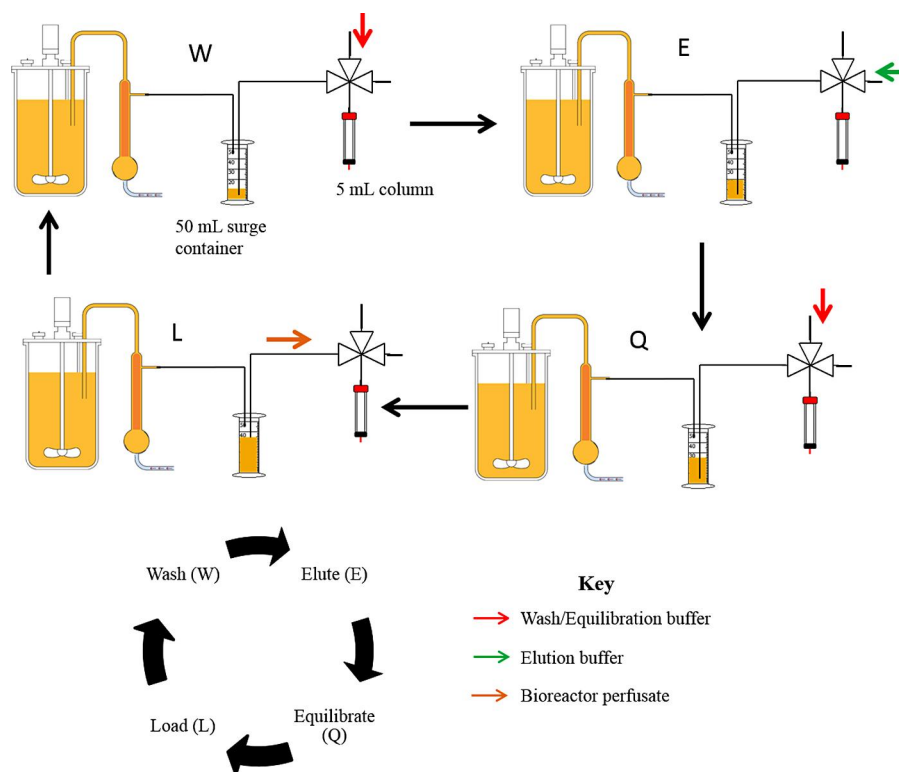


Figure 3. Schematic of process synchronization in OCC. Purification begins with about 20 mL perfusate in the holding bottle. The cycle begins at the end of a load (L) cycle and a beginning of the wash (W) cycle. The wash cycle is done with 10 column volumes for 10 min (13.9-mL collected in holding bottle). After wash, there is elution with 5 column volumes for 5 min (20.8 mL total perfusate collected). The equilibration step takes another 10 column volumes for 10 more minutes (34.7 mL total collected in holding bottle). The load step takes 10 min to load 50 mL of perfusate (13.9 + 34.7 mL = 48.6 mL total collected). The excess load over perfused is compensated during the 20 min sanitization steps that occur once every 20 cycles. *Note:* OCC, one-column continuous chromatography.

Implementation of SMB or PCC requires acquisition of new and expensive equipment which may be a financial burden to smaller organizations. In addition to equipment costs, it requires extensive facility modifications to be able to fully use the new equipment. The current approach requires little to no facility modification although a detailed process understanding is required which might involve process development and optimization.

Process monitoring

The proposed method simplifies the process of accurately monitoring column performance and making column switching decisions. In SMB and PCC, there are multiple columns, each requiring continuous monitoring. Even with the use of multivariate statistics, this becomes cumbersome. Industry practice involves the use of pulse-input height equivalent of a theoretical plates (HETP),^[15,16] monitoring of critical parameters such as step yield and qualitative review of chromatographic data by trained personnel. The downside with pulse-input HETP in integrated continuous biomanufacturing is that it requires stopping the process for testing. This becomes impractical since perfect flow synchronization will eventually result in process mismatch. Moreover, relying on cycle number provided by the manufacturer might be unreliable and does not ascribe to quality by design (QbD) and process analytical technology (PAT) principles. In OCC, the use of a single column ensures the development of a

detailed column monitoring regime, providing detailed information on column performance and life cycle for ensuring accurate column switching decisions. In a future paper, the authors describe the application of PAT for monitoring column performance in real time and making accurate column switching decisions. Models relating process information such as multiwavelength UV absorbance, conductivity, pH, and cycle volumes to critical information such as product step yield and purity will be discussed.

Results

For testing purposes, a therapeutic monoclonal antibody (MAb) was produced from in-house CHO cells under

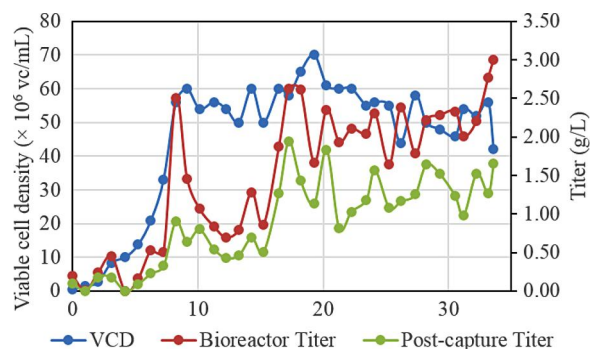


Figure 4. Cell density and titer of daily crude unpurified bioreactor sample and affinity chromatography purified sample as a function of culture days.

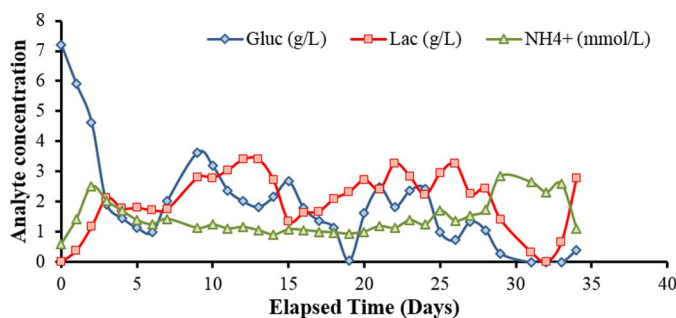


Figure 5. Nutrient and waste control in bioreactor over 35-day perfusion bioreactor run.

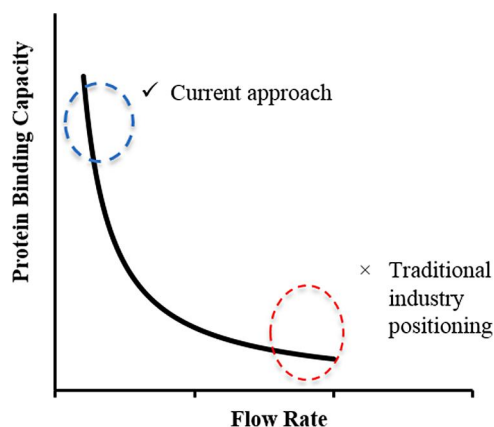


Figure 6. Generic variation of binding capacity as a function of flow rates. This shows the superiority of current approach in optimizing dynamic binding capacity over current industry practices.

perfusion in a 2 L bioreactor for 35 days as described in the Materials and Methods section. The growth profile, daily bioreactor titer, and daily pooled protein A purified sample values are shown in Figure 4, while the glucose, ammonia, and lactate profiles are shown in Figure 5. Media optimization techniques allowed for maintaining perfusion rates at one vessel volume/day on average throughout the run.

Column operations were optimized to ensure that the total time for a single cycle was equivalent to the rate of production of perfusate from the reactor: a single-column operation with 50 mL load volume took 35 min (rate of perfusion 48.6 mL/35 min).

Figure 6 shows a generic variation of binding capacity with flow rate and illustrates the clear advantage of operating chromatography processes under low flow rates to optimize resin utilization.

During multicycle column chromatography, there are several observable changes in the chromatogram. On the absorbance spectra, as the column degrades, the elution peak broadens in addition to tailing, the maximum absorbance of the elution peak decreases (as shown in Figure 7a), and the cycle yield decreases (shown by a decrease in area under the peak). Another observation is the increase in absorbance of the load peak. As the protein A resin degrades, fewer binding sites are available, resulting in an increase in antibody flow-through, and thus an increase in the load peak. The higher flow-through in turn results in a decreased amount of bound antibody resulting in a decrease in cycle yield.

When elution peak height was more than one standard deviation from the average, the column was switched as shown in Figure 7b. The extra cycles were conducted in this case to demonstrate the irreversible nature of this parameter deviation.

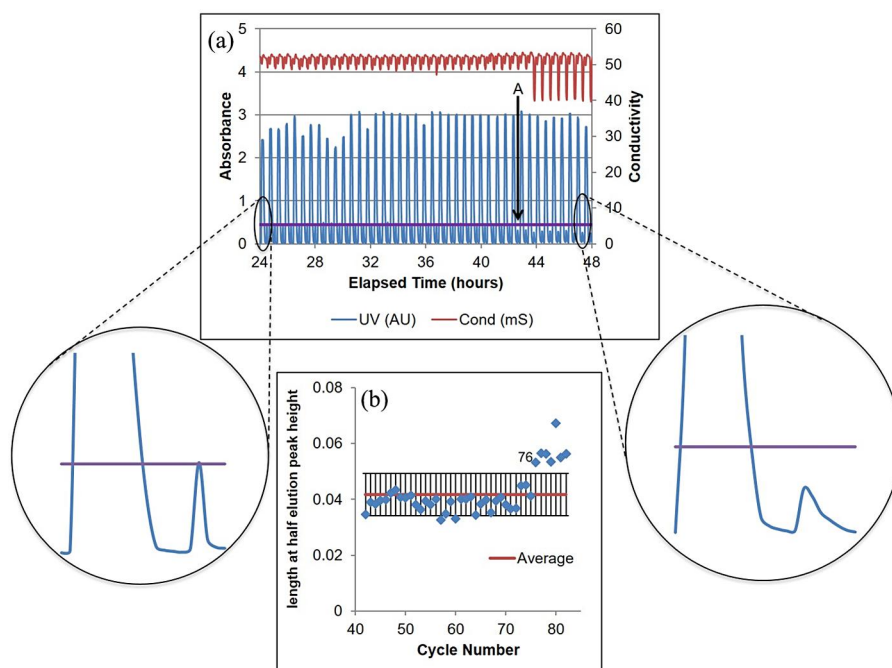


Figure 7. (a) UV and conductivity profiles during continuous antibody capture. Only last 24 hr prior to column failure is shown to highlight the consistent column performance and the observable column failure. Point A indicates the point of observable deviation of elution peak maximum from previous values. (b) Peak length at half elution peak height for last 40 cycles in single prepacked column.

Figure 7a also shows the last 24 hr prior to column failure. The figure shows that by visual observation, the maximum absorbance value of the elution peak deviated from the previously almost constant value after about 73 cycles (42 hr including sanitization/regeneration steps). However, the simplified approach used in this study showed a significant deviation outside the set limits three cycles later.

Another important advantage of this approach is the very low level of HCP in the bioreactor and purified material. Since the bioreactor is operating under perfusion, which averts accumulation, there are already low levels of HCP in the perfusate. In this study, triplicate samples from the bioreactor and from the purified product were taken and analyzed for HCP content, and the average values were found to be 55 and 8.5 ppm respectively, much lower than the acceptable limits in drug products.

Discussion

By controlling media components, the levels of critical nutrients in the bioreactor were kept at adequate levels. This tight control of nutrients ensured production of material within acceptable product quality requirements.

An important component of the current approach as mentioned earlier is the strict method synchronization. Previous methods had set perfusion rates as a function of cell density.^[13] This approach could not be used in the current study because perfusion rates will change as a function of cell density disrupting method synchronicity. To resolve the increasing nutrient needs as cell density increased, process development approaches were used to optimize critical nutrient composition in media. This resulted in in-house media formulation efficient for growth and production phases: one media for growth and another for production. This offered a significant advantage in terms of process simplicity and makes future scale-up efforts less cumbersome.

Perfusion media used during cell culture was stored at 4°C in darkness in cell culture bags. Cell culture bags are currently validated for only 30 days due to the increase in leachates. Hence, preparing two media batches with different nutrient compositions was beneficial, given the limitations of current disposable technology. To maintain the sterile barrier, a manifold containing two inlet streams to the media feed line was prepared during bioreactor assembly. During the production phase, the feed line of the growth media was simply clamped and the new media containing production phase was welded under sterile conditions. This approach can be easily implemented under cGMP settings.

The lower perfusion rates result in higher antibody concentration in the perfusate. Nonetheless, the lower flow rates are preferred for dynamic binding capacity.^[17,18] From Figure 6, it is clear that binding capacity has an inverse relationship to flow rate, thus as flow rate decreases, the binding capacity increases. In batch and fed-batch processes, large volumes of harvest have to be processed in as little time as possible, resulting in high flow rates. The lower flow rates in this approach presents an important advantage: it has been found that at higher titers, a shorter residence time results in a significant decrease in dynamic binding capacities.^[19] Consequently, as

the continuous upstream method is constantly improved to give higher titers, the current approach of operating at low flow rates will maintain longer residence times and hence prove a robust method for the continuous capture of the therapeutic antibodies compared to the higher flow rate approaches.

On the other hand, the continuous harvesting of therapeutic antibodies from a cell culture bioreactor will limit the extent to which concentrations can increase within the bioreactor compared to batch and fed-batch processes. This is relevant because it has been observed that for superficial velocities up to 6 cm/min, there was a minimal change in fraction of feed bound to the column.^[20] Considering the current setup in this study, even for a 20-fold increase in flow rates (unsupported by the column), there is still not a significant increase in superficial velocity to affect binding capacity. Thus, the dual impact of relatively lower bioreactor titers and significantly lower superficial velocities will result in very high dynamic binding capacities with sufficient flexibility to make process optimization.

With regard to column capacity, the MabSelect columns used in this study have a rated dynamic binding capacity at 10% breakthrough of 30 mg/mL for human IgG. Thus, using the same setup (5 mL prepacked affinity column) and operating conditions proposed in this study, a bioreactor antibody concentration of up to 3 mg/mL can still be conveniently supported. Moreover, even as antibody concentrations improve with better perfusion process understanding and control, simply increasing column size and optimizing purification cycles would yield similar result. This ensures a fairly robust and transferable platform.

The current approach proposes one column cyclic chromatography in contrast to multicolumn approaches for use in an integrated system. The comparison of productivity and capacity utilization between the suggested system and multicolumn systems has been evaluated.^[21] The authors found that by running these cyclic processes, capacity utilization tends toward 100% at low flow rates, which is the suggested approach in this study. This is an important aspect of the current approach. In addition to this, the authors found a ~ 27% loss in productivity in the batch compared to the multicolumn system. It is important to note that productivity can be improved by optimization experiments such as load strategy.^[22] Thus, implementation of multicolumn operation is only one method for improving resin productivity. In this study, the productivity of the affinity resin was improved to 34 mg antibody/mL-resin.hr compared to the 19.7 mg antibody/mL-resin.hr previously observed.^[21] Moreover, simulations have found better performance for a single-column system compared to a multicolumn countercurrent system with only a slight improvement in productivity.^[23]

Consider the OCC approach proposed in this work producing an average of 1.25 g/L/day of purified antibody (Figure 4) compared to a best-case scenario 10-day fed-batch process producing 5 g/L crude followed by a 1-day affinity chromatography purification process achieving 90% yield. Table 1 summarizes the time advantage for the production of 1 kg of affinity-purified antibody using the OCC approach proposed in this work compared to the high titer fed-batch process.

Table 1. Comparison of OCC approach vs traditional fed-batch for the production of 1 kg of affinity purified antibody.

	OCC process	Traditional fed-batch
Reactor volume (L)	50	50
Culture duration (days)	Continuous	10
Purification time (days)	Continuous	1
Time to produce 1 kg of affinity purified antibody (days)	26 [†]	44 [†]

[†]Data reported in this work. 10 days to reach peak cell density and 16 days of production (1.25 g/L/day affinity purified).

[‡]Assuming 5 g/L titer at harvest with 90% yield for affinity chromatography.

As seen in Table 1, for the same reactor volume, it will take 44 days (4 runs) to produce the same amount of antibody compared to only 26 days on the perfusion-based OCC approach.

Another important aspect of the current approach is the column switching strategy. Prepacked MabSelect columns are usually rated to 100 cycles. However, to properly align with PAT and QbD, there has to be a rigorous monitoring technique, designed to detect and even predict when the protein A column begins to fail. In the current OCC approach, upstream and downstream synchronizations are so tightly controlled that traditional pulse testing approaches are not practical.

Transition analysis has been proposed as a column monitoring tool for estimating the overall column integrity value in repeated column operations.^[24] However, in the current study, since the resin quantity used had been significantly reduced already, and current hardware could not be easily harmonized with a real-time integrity value calculator, a simplified approach was taken for making column switching decisions.

The elution peak was assumed to be an isosceles triangle with the peak height, the triangle height, and the integration of the area under the peak taken to be the triangle area. Using these two values, the peak length at half the peak height can be estimated. As the column degrades, it is expected that the peak will broaden and this length will increase. Although this approach is simplistic, it provides an estimate of the decrease in cycle yield, elution peak height, and peak broadening. These are all factors that indicate column degradation.

A real-time analysis of elution peak length at half the height was able to predict column failure within three cycles to that seen by a detailed analysis of spectral data by trained personnel. This objective, yet simple method of column monitoring, presents a very practical method for making column switching decisions. The OCC approach can eventually be easily implemented regardless of existing infrastructure. This approach is useful as a qualitative tool but requires some improvement to be considered as an acceptable tool in manufacturing settings.

The authors are currently working on an expanded integrity value for the column that considers not only single wave absorbance data but also multiwave absorbance in addition to conductivity and pH data. The goal will be to provide more accurate data on column performance, with regard to step yield and purity. This information will be very valuable in larger manufacturing settings with large expensive columns where the current simplistic approach might be too costly.

Of significance, is the fact that the one column approach in this study equally reduces the number of failure modes? For instance, with SMB, monitoring all six columns remains very cumbersome and the failure of any one of the columns will result in a disruption in the process before the faulty column can be removed and replaced. The presence of so many columns increases the odds of repeated failures during long operations (30–90 days) that are common in currently proposed integrated bioprocesses.

Affinity chromatography is an excellent method for eliminating the bulk HCP from the perfusate.^[25,26] The industry HCP content typically allowed in antibody products is 100 ppm.^[27,28] Thus, the observed HCP content from the current study is so low that even with a fivefold concentration protein A purified material for next step purification, the HCP concentration will still be within allowable limits. This makes the current approach an excellent candidate for first-in-human studies valuable to smaller research and manufacturing companies and academic research institutes with limited financial resources.

Summary and conclusion

The OCC approach as a technological platform for integrated continuous biomanufacturing process offers multiple advantages over previously proposed integrated continuous processes in terms of cost savings, ease of implementation, and flexibility. In this study, a first successful case has been presented. The setup has been used to successfully grow and capture a MAb over an extended period of time. This platform provides an efficient and cost-effective approach for implementing integrated continuous biomanufacturing for the production of a wide range of therapeutic proteins.

In this work, perfusion rates upstream were reduced to 1 vv/d by performing cell culture medium optimization. The result was two optimized media: one for growth and the other for production. The two optimized media simplified process operations that will eventually lead to less disruptive implementation in manufacturing settings. This low perfusion rate ensured that the bioreactor perfusate was at sufficiently high concentrations. It equally reduced the volume of material produced for downstream processing.

The combined effect of higher titers and smaller processing volumes permitted the use of a small affinity column (<1% of reactor volume), low superficial velocities, and high residence times downstream. These downstream operating conditions optimized protein A use, resulting in optimal dynamic binding capacities and leading to near 100% capacity utilization.

In addition to this, the platform can be implemented in diverse manufacturing environments with existing equipment without the need to purchase new costly equipment that might require facility modifications. This is particularly relevant for application in biosimilars programs and early phase process development which still have high failure rates. Validation of the current approach at the pilot scale using a 50 L bioreactor operating in continuous mode over 60 days for the manufacture of sufficient material for first-in-human studies will provide useful data for industry professionals exploring innovative approaches to implementing integrated continuous

biomanufacturing in current and future products. This platform offers an attractive option for cGMP manufacturing, especially for small companies, start-ups, and academic research institutes with limited resources. The compelling economic benefits of this approach could potentially encourage industry professionals to make necessary investments which would without doubt propel the industry to a new and more profitable future.

Conflict of interest

The authors have no conflict of interest to disclose.

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References

- [1] Schaber, S. D.; Gerogiorgis, D. I.; Ramachandran, R.; Evans, J. M. B.; Barton, P. I.; Trout, B. L. Economic Analysis of Integrated Continuous and Batch Pharmaceutical Manufacturing: A Case Study. *Ind. Eng. Chem. Res.* **2011**, *50*(17), 10083–10092. DOI: [10.1021/ie2006752](https://doi.org/10.1021/ie2006752).
- [2] Walther, J.; Godawat, R.; Hwang, C.; Abe, Y.; Sinclair, A.; Konstantinov, K. The Business Impact of an Integrated Continuous Biomanufacturing Platform for Recombinant Protein Production. *J. Biotechnol.* **2015**, *213*, 3–12. DOI: [10.1016/j.jbiotec.2015.05.010](https://doi.org/10.1016/j.jbiotec.2015.05.010).
- [3] Pollock, J.; Ho, S. V.; Farid, S. S. Fed-Batch and Perfusion Culture Processes: Economic, Environmental, and Operational Feasibility Under Uncertainty. *Biotechnol. Bioeng.* **2013**, *110*(1), 206–219. DOI: [10.1002/bit.24608](https://doi.org/10.1002/bit.24608).
- [4] Konstantinov, K. B.; Cooney, C. L. White Paper on Continuous Bioprocessing. May 20–21, 2014 Continuous Manufacturing Symposium. *J. Pharm. Sci.* **2015**, *104*(3), 813–820. DOI: [10.1002/jps.24268](https://doi.org/10.1002/jps.24268).
- [5] Xenopoulos, A. A New, Integrated, Continuous Purification Process Template for Monoclonal Antibodies: Process Modeling and Cost of Goods Studies. *J. Biotechnol.* **2015**, *213*, 42–53. DOI: [10.1016/j.jbiotec.2015.04.020](https://doi.org/10.1016/j.jbiotec.2015.04.020).
- [6] Zydney, A. L. Perspectives on Integrated Continuous Bioprocessing—Opportunities and Challenges. *Curr. Opin. Chem. Eng.* **2015**, *10*, 8–13. DOI: [10.1016/j.coche.2015.07.005](https://doi.org/10.1016/j.coche.2015.07.005).
- [7] Ozturk, S. S. Opportunities and Challenges for the Implementation of Continuous Processing in Biomanufacturing. In *Continuous Processing in Pharmaceutical Manufacturing*; Subramanian, G., Ed.; John Wiley & Sons, Weinheim, Germany, 2014; pp 457.
- [8] Birch, J. R.; Racher, A. J. Antibody Production. *Adv. Drug. Deliv. Rev.* **2006**, *58*(5–6), 671–685. DOI: [10.1016/j.addr.2005.12.006](https://doi.org/10.1016/j.addr.2005.12.006).
- [9] Kefler, L. C.; Gueorgieva, L.; Rinas, U.; Seidel-Morgenstern, A. Step Gradients in 3-Zone Simulated Moving Bed Chromatography. Application to the Purification of Antibodies and Bone Morphogenetic Protein-2. *J. Chromatogr. A.* **2007**, *1176*(1–2), 69–78.
- [10] Seidel-Morgenstern, A.; Kefler, L. C.; Kaspereit, M. New Developments in Simulated Moving Bed Chromatography. *Chem. Eng. Technol.* **2008**, *31*(6), 826–837. DOI: [10.1002/ceat.200800081](https://doi.org/10.1002/ceat.200800081).
- [11] Girard, V.; Hilbold, N. J.; Ng, C. K.; Pegon, L.; Chahim, W.; Rousset, F.; Monchois, V. Large-Scale Monoclonal Antibody Purification by Continuous Chromatography, from Process Design to Scale-Up. *J. Biotechnol.* **2015**, *213*, 65–73. DOI: [10.1016/j.jbiotec.2015.04.026](https://doi.org/10.1016/j.jbiotec.2015.04.026).
- [12] Ng, C. K. S.; Rousset, F.; Valery, E.; Bracewell, D. G.; Sorensen, E. Design of High Productivity Sequential Multi-Column Chromatography for Antibody Capture. *Food Bioprod. Process.* **2014**, *92*(2), 233–241. DOI: [10.1016/j.fbp.2013.10.003](https://doi.org/10.1016/j.fbp.2013.10.003).
- [13] Warikoo, V.; Godawat, R.; Brower, K.; Jain, S.; Cummings, D.; Simons, E.; Johnson, T.; Walther, J.; Yu, M.; Wright, B.; et al. Integrated Continuous Production of Recombinant Therapeutic Proteins. *Biotechnol. Bioeng.* **2012**, *109*(12), 3018–3029. DOI: [10.1002/bit.24584](https://doi.org/10.1002/bit.24584).
- [14] Angarita, M.; Muller-Spath, T.; Baur, D.; Lievrouw, R.; Lissens, G.; Morbidelli, M. Twin-Column CaptureSMB: A Novel Cyclic Process for Protein A Affinity Chromatography. *J. Chromatogr. A.* **2015**, *1389*, 85–95. DOI: [10.1016/j.chroma.2015.02.046](https://doi.org/10.1016/j.chroma.2015.02.046).
- [15] Hofmann, M. A Novel Technology for Packing and Unpacking Pilot and Production Scale Columns. *J. Chromatogr. A.* **1998**, *796*(1), 75–80. DOI: [10.1016/S0021-9673\(97\)00727-9](https://doi.org/10.1016/S0021-9673(97)00727-9).
- [16] Moscardiello, J.; Purdom, G.; Coffman, J.; Root, T. W.; Lightfoot, E. N. Characterizing the Performance of Industrial-Scale Columns. *J. Chromatogr. A.* **2001**, *908*(1–2), 131–141. DOI: [10.1016/S0021-9673\(00\)01062-1](https://doi.org/10.1016/S0021-9673(00)01062-1).
- [17] Hober, S.; Nord, K.; Linholt, M. Protein A Chromatography for Antibody Purification. *J. Chromatogr. B Analyt. Technol. Biomed. Life Sci.* **2007**, *848*(1), 40–47. DOI: [10.1016/j.jchromb.2006.09.030](https://doi.org/10.1016/j.jchromb.2006.09.030).
- [18] Aldington, S.; Bonnerjea, J. Scale-Up of Monoclonal Antibody Purification Processes. *J. Chromatogr. B.* **2007**, *848*(1), 64–78. DOI: [10.1016/j.jchromb.2006.11.032](https://doi.org/10.1016/j.jchromb.2006.11.032).
- [19] Natarajan, V.; Zydney, A. L. Protein A Chromatography at High Titrers. *Biotechnol. Bioeng.* **2013**, *110*(9), 2445–2451. DOI: [10.1002/bit.24902](https://doi.org/10.1002/bit.24902).
- [20] Ghosh, R. Separation of Proteins Using Hydrophobic Interaction Membrane Chromatography. *J. Chromatogr. A.* **2001**, *923*(1–2), 59–64. DOI: [10.1016/S0021-9673\(01\)00956-6](https://doi.org/10.1016/S0021-9673(01)00956-6).
- [21] Baur, D.; Angarita, M.; Muller-Spath, T.; Morbidelli, M. Optimal Model-Based Design of the Twin-Column Capture SMB Process Improves Capacity Utilization and Productivity in Protein A Affinity Capture. *Biotechnol. J.* **2016**, *11*(1), 135–145. DOI: [10.1002/biot.201500223](https://doi.org/10.1002/biot.201500223).
- [22] Ghose, S.; Nagrath, D.; Hubbard, B.; Brooks, C.; Cramer, S. M. Use and Optimization of a Dual-Flowrate Loading Strategy To Maximize Throughput in Protein-A Affinity Chromatography. *Biotechnol. Prog.* **2004**, *20*(3), 830–840. DOI: [10.1021/bp0342654](https://doi.org/10.1021/bp0342654).
- [23] Bochenek, R.; Marek, W.; Piatkowski, W.; Antos, D. Evaluating the Performance of Different Multicolumn Setups for Chromatographic Separation of Proteins on Hydrophobic Interaction Chromatography Media by a Numerical Study. *J. Chromatogr. A.* **2013**, *1301*, 60–72. DOI: [10.1016/j.chroma.2013.05.053](https://doi.org/10.1016/j.chroma.2013.05.053).
- [24] Larson, T. M.; Davis, J.; Lam, H.; Cacia, J. Use of Process Data To Assess Chromatographic Performance in Production-Scale Protein Purification Columns. *Biotechnol. Prog.* **2003**, *19*(2), 485–492. DOI: [10.1021/bp025639g](https://doi.org/10.1021/bp025639g).
- [25] Hogwood, C. E.; Tait, A. S.; Koloteva-Levine, N.; Bracewell, D. G.; Smales, C. M. The Dynamics of the CHO Host Cell Protein Profile during Clarification and Protein A Capture in a Platform Antibody Purification Process. *Biotechnol. Bioeng.* **2013**, *110*(1), 240–251. DOI: [10.1002/bit.24607](https://doi.org/10.1002/bit.24607).
- [26] Nian, R.; Zhang, W.; Tan, L.; Lee, J.; Bi, X.; Yang, Y.; Gan, H. T.; Gagnon, P. Advance Chromatin Extraction Improves Capture Performance of Protein A Affinity Chromatography. *J. Chromatogr. A.* **2016**, *1431*, 1–7. DOI: [10.1016/j.chroma.2015.12.044](https://doi.org/10.1016/j.chroma.2015.12.044).
- [27] Champion, K. M.; Nishihara, J. C.; Joly, J. C.; Arnott, D. Similarity of the *Escherichia coli* Proteome upon Completion of Different Biopharmaceutical fermentation Processes. *Proteomics.* **2001**, *1*(9), 1133–1148. DOI: [10.1002/1615-9861\(200109\)1:9<1133::aid-prot1133>3.3.co;2-j](https://doi.org/10.1002/1615-9861(200109)1:9<1133::aid-prot1133>3.3.co;2-j).
- [28] Wang, X.; Hunter, A.; Mozier, N. M. Host Cell Proteins in Biologics Development: Identification, Quantitation and Risk Assessment. *Biotechnol. Bioeng.* **2009**, *103*(3), 446–458. DOI: [10.1002/bit.22304](https://doi.org/10.1002/bit.22304).